

Question Option A				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
6	(a)		(i)	A disease which is always present at low levels (in an area)/frequently at a predictable rate in a specific location/population (1) Epidemic – {significant/large} increase in the usual number of cases/rapid spread of infectious disease to a large number of people within a short period (1)	2			2		
			(ii)	sewage entered water supply (1) Cholera spread by drinking contaminated water/feco-oral route (1)	1	1		2		
			(iii)	Carriers/someone with the disease brought into Haiti			1	1		
	(b)		(i)	Antibodies are specific to an antigen (1) Different Strains would have different antigens (1) If no agglutination they different strains/if tested using antibodies to O1, O139 would not show agglutination/OR(1)	1	2		3		
			(ii)	Identify the strain of <i>V.cholerae</i> in the peacekeepers in Haiti (1) Compare distribution of known strains in the {world/Nepal} to locate possible source (1)			2	2		2
			(iii)	Any three (x1)from : Antibiotics will pass through the gut before all bacteria killed (1) <i>V. cholerae</i> is Gram-negative and some antibiotics {less/not} effective/narrow spectrum (1) Kill bacteria but toxin remains (1) Antibiotic resistance (1)		1	2	3		
			(iv)	583/583.3(cm ³) 2 marks $\frac{200}{24} \times 70 = 1$ mark		2		2	2	
	(c)		(i)	(First dose-) for a primary (immune) response and (Second dose) – for a secondary(immune) response (1) second dose acting as a booster/to increase antibody levels/increase memory cells (1)	2			2		
			(ii)	The vaccine would pass through the digestive system/ the vaccine would be in the intestine long enough/not {enough/all} absorbed/ vaccine could be broken down / stomach acid stops acid working/owtte		1		1		